

FELIPE RODRIGUES PEIXOTO DA SILVA

Full-Stack Developer · Applied AI

São Paulo, Brazil · frps2003@gmail.com · github.com/Comunalti

SUMMARY

Full-stack developer with 3+ years in industry building complete web platforms: APIs in Node and Python, relational and non-relational databases, message queues, containers and interfaces in React and Flutter. Computer Engineer (Instituto Mauá de Tecnologia, 2025). Code guided by SOLID, DRY and design patterns, focused on UI-administrable systems that reduce operational dependency on developers. Beyond code: two years teaching best practices as a scholarship TA and a term as student organization president, leading multidisciplinary teams.

PROFESSIONAL EXPERIENCE

Full-Stack Developer · Intelicity Dec 2024 — present

- Single-handedly delivered, end to end (data modeling, APIs, frontend, dockerization and deployment), an urban roadwork management platform for city governments: granular access control (users, roles, per-page and per-feature permissions) and event-triggered automations — all configurable through the admin interface, with zero code changes.
- Rebuilt the data-annotation platform with Flutter, FastAPI and MongoDB: parallel image downloads, a single flexible workflow and 100% UI-based administration, removing developer dependency from operations.
- Worked with the AI team on migrating image processing to a distributed architecture: containerized services, orchestration and RabbitMQ messaging.

Development Intern — AI · Intelicity Mar 2023 — Dec 2024

- Computer vision pipelines in Python: object detection and classification across large volumes of images, with georeferenced data in PostgreSQL.
- Identified the bottleneck in the manual data-annotation workflow and proposed SGC (General Classification System): a C# application with per-user authentication and real-time API synchronization — adopted by the operations team as the official tool.

HIGHLIGHTED PROJECTS

Data-driven software architecture (Thesis) — A C# engine that interprets rules declared in an external XML file and assembles entities at runtime, with zero recompilation: serialized state machine, declarative parsing and a custom visual editor — the same principles behind enterprise systems extensible through configuration.

Personal full-stack website — React + Node + TypeScript on Docker Compose, bilingual API-served content, route-based presentation variants (formal/creative, with light/dark theme) and continuous deployment (Vercel + Render) with a custom domain.

EDUCATION

B.Sc. in Computer Engineering · Instituto Mauá de Tecnologia 2021 — 2025

Mechatronics Technician (integrated with high school) · ETEC Júlio de Mesquita 2018 — 2020

ACADEMIC ACTIVITIES

- Elected president of a software development student organization (2024; member since 2022): led multidisciplinary teams — programming, art and production — with cycle planning, level-based delegation, deadline tracking and an organized succession when stepping down.
- Scholarship teaching assistant for two consecutive years (2023–2024): taught SOLID, DRY and design patterns (Gang of Four) to younger students, reviewing their projects and lecturing on computer graphics and GPU programming — the foundation of my mentoring and code review practice.

HARD SKILLS

Languages	TypeScript/JavaScript, Python, C#, Dart, SQL
Backend	Node.js, Express, FastAPI, REST APIs, RabbitMQ, WebSockets
Frontend	React, Vite, Flutter, HTML/CSS
Data	PostgreSQL (incl. georeferenced data), MongoDB
DevOps	Docker, Docker Compose, container orchestration, CI/CD, Linux, Git
AI & Vision	Computer vision, object detection/classification, data annotation
Architecture & Practices	Distributed systems, messaging, RBAC, SOLID, DRY, design patterns (GoF)
Other	Unity, GPGPU/compute shaders, procedural generation

SOFT SKILLS

People management and team leadership

Technical mentoring and teaching (2 years as TA)

End-to-end ownership (entire platform built solo)

Cross-disciplinary collaboration

Clear technical communication

Pragmatic problem solving

Continuous learning

Product thinking for non-technical users

LANGUAGES

Portuguese — native English — professional (technical reading, writing and documentation)